ENHANCING ROBUSTNESS OF DEEP NETWORKS BASED ON A TWO-PHASE MODEL OF THEIR TRAINING WITH NOISY LABELS

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ABSTRACT

In this study we model explicitly the learning behavior of deep neural networks (DNNs) trained with noisy labels in image classification. Specifically, we show theoretically and experimentally that the training process can be divided into two phases: a *learning phase* in which the outputs of DNNs converge to a hidden noise distribution no matter whether the training samples are clean or noisy; and a *memorization phase* in which DNNs start to overfit until the output for each training sample converges to its corresponding noisy one-hot labels. This two-phase model enables us to develop two simple yet accurate methods that rely on the outputs of DNNs to estimate the noise transition matrix (NTM). It also enables us to resolve a pitfall of many existing methods for robust training under noisy labels based on the *small-loss assumption*, namely clean samples have smaller loss than noisy samples in the early training phase. We show that these methods fail when NTM is not a column diagonally-maximum matrix and that this pitfall can be fixed by modifying the small-loss assumption based on our NTM estimation methods.

1 INTRODUCTION

Deep neural networks (DNNs) have shown remarkable generalization ability in different domains, such as image classification (Krizhevsky et al., 2012; He et al., 2016), semantic segmentation (Lin et al., 2016; Long et al., 2015) and others. However, DNNs also generalize poorly by memorizing the whole training dataset under the noisy labels due to the over-parameterized of the model (Zhang et al., 2021). Carefully characterizing the training process under noisy labels not only helps us interpret the learning behavior of DNNs, but also can inspire researchers to improve the model generalization by designing different robust training strategies in the presence of label noises.

One of the popular strategies to improve generalization of DNNs is to model the hidden noise structure, which is usually formulated by the noise transition matrix (NTM). The typical method is adding an extra noise layer after the softmax layer and adapting the model predictions to match the noisy label distributions, where the weights of the noise layer are the estimation of NTM (Sukhbaatar et al., 2014). By simply removing the noisy layer, one can easily reach the noise-free prediction of DNNs. Similar strategies include adding a separate network (Han et al., 2018a; Yao et al., 2018) or explicit estimate NTM by EM algorithms (Bekker & Goldberger, 2016).

Another popular strategy for handling label noises is to filter out the noisy samples during the training process. Based on the learning behavior that DNNs prior learn simple patterns first then memorize noisy samples (Arpit et al., 2017), many existing methods assumed that clean samples will have smaller loss compared to noisy ones and designed different strategies to iteratively filter out larger loss samples at the early training epochs (Han et al., 2018b; Yu et al., 2019; Wei et al., 2020). We summarize these methods are based on the *small-loss assumption* (formally definition of this assumption is shown in Section 2.2) and demonstrate that these methods cannot handle noisy labels under a specific scenario.

In this paper, we make three key contributions compared to prior work on exploring the learning behavior of DNNs trained with noisy labels.

- We model explicitly the training process into a *learning phase* and a *memorization phase*. We provide theoretical proof and experimental demonstration that the outputs of DNNs first converge to the noise distributions in the learning phase and then converge to the corresponding noisy one-hot labels in the memorization phase (Section 3). Inspired by this finding, we focus on two critical challenges in learning with noisy labels.
- *Challenge 1: Estimation of the noise transition matrix (NTM).* By applying a clean validation set, we propose two simple yet accurate methods that only rely on the outputs of DNNs to estimate NTM (Section 4).
- *Challenge 2: Robust training DNNs under noisy labels.* We theoretically and experimentally demonstrate that existing DNNs based on the small-loss assumption fail in image classification when NTM is not a column diagonally-maximum matrix. However, we fix this pitfall by simply modifying the small-loss assumption based on our NTM estimation methods. (Section 5).

2 PRELIMINARIES

2.1 NOTATION

Scenario without label noise. In c-class classification, let $[c] = \{1, 2, ..., c\}$ for the set of c unique class labels. Given n training data $X = (x, y^*)^n \in (\mathbb{R}^d, [c])^n$ where $x \in \mathbb{R}^d$ denotes the input images and y^* denotes the clean labels $y^* \in [c]$. If label y^* is given as i class, y^* can be represented by a one-hot vector e^i where i^{th} component of e^i equals 1 and others equal 0.

For a classification model $f(y^*|x, \theta)$ parameterized by θ, we denote the *softmax* output of $f(y^*|x, \theta)$ as $\hat{p}_\theta(y^*|x)$. The training process can be formulated as the search of the model $f \in \mathcal{F}$ that minimize the empirical risk of a given *cross-entropy* loss $\ell(\hat{p}_\theta(y^*|x), y^*)$:

$$\ell(\hat{p}_\theta(y^*|x), y^* = e^i) = -(e^i)^T \log \hat{p}_\theta(y^*|x) = -\log \hat{p}_\theta(y^* = i|x) \tag{1}$$

Scenario with label noise. Based on the class-conditional classification noise process (Angluin & Laird, 1988), we assume that every *observed* noisy labels $\widetilde{y}$ in class $j \in [c]$ is independently flipped from an *unobserved* clean label y^* in class $i \in [c]$ with probability $p(\widetilde{y} = j|y^* = i)$. We abbreviated $p(\widetilde{y} = j|y^* = i)$ as q_{ji}. We use subscript of X to denote the subset in training dataset X, e.g., $X_{\widetilde{y}=j,y^*=i}$ denotes the samples which incorrectly annotate class i as class j in X. The noisy label distribution of $\widetilde{y}$ given by q_{ji} can be parameterized by a $c \times c$ probability transition matrix Q, where $Q_{ji} = q_{ji}, \forall i, \forall j$. We name matrix Q as the noise transition matrix (NTM). The j^{th} row and i^{th} column in Q are denoted as Q_{j*} and Q_{*i}, respectively. Examples of NTM are shown in Figure 1.

2.2 DEFINITIONS

We pre-list some of the important definitions that will be used in the following sections.

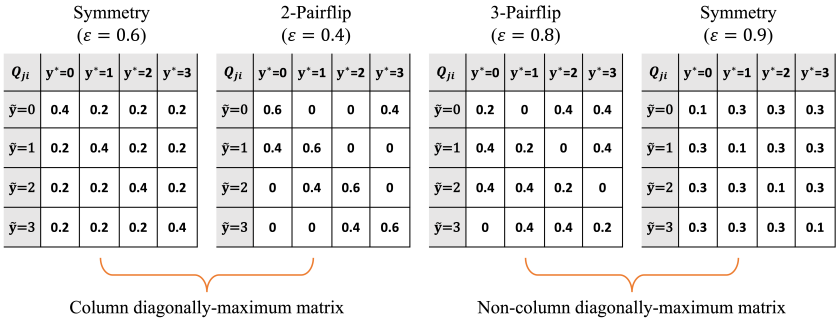
Definition 1 (*K-Pairflip noisy label*): Each hidden clean label y^* is independently flipped to K other classes with the noise ratio ε. Formally, in NTM, $\forall i \in [c]$, $q_{ii} = 1 - \varepsilon$, $\sum_{k \in K} q_{ki} = \varepsilon$ for some $k \neq i$ and $q_{ji} = 0$ otherwise. Example NTM of *K-Pariflip* noisy labels are shown in Figure 1.

Definition 2 (*Symmetry noisy label*): Each hidden clean label y^* is independently and uniformly flipped to all other classes with the noise ratio ε. Formally, in NTM, $\forall i \in [c]$, $q_{ii} = 1 - \varepsilon$ and $q_{ji} = \frac{\varepsilon}{(c-1)}, \forall j \neq i$. Example NTM of *Symmetry* noisy labels are shown in Figure 1.

Definition 3 (*Prediction confusion matrix*): For the softmax outputs of DNNs $\hat{p}_\theta$, we define the prediction confusion matrix $M = \{m_{ji}\}$ by $m_{ji} = \frac{\sum_{x \in X_{y^*=i}} \hat{p}_\theta(\widetilde{y}=j|y^*=i,x)}{|X_{y^*=i}|}$.

Definition 4 (*Column diagonally-maximum matrix*): We name the matrix $Q = \{q_{ji}\}$ column diagonally-maximum if $q_{ii} > q_{ji}, \forall i, i \neq j$, indicating the maximum value of each column is located on the diagonal of Q. Examples are shown in Figure 1.

Definition 5 (*Small-loss assumption*): Based on the behavior that DNNs learn simple patterns first (Arpit et al., 2017), the small loss sample more likely has a clean label in the early training phase.



Q_{ji}	$y^*=0$	$y^*=1$	$y^*=2$	$y^*=3$
$\tilde{y}=0$	0.4	0.2	0.2	0.2
$\tilde{y}=1$	0.2	0.4	0.2	0.2
$\tilde{y}=2$	0.2	0.2	0.4	0.2
$\tilde{y}=3$	0.2	0.2	0.2	0.4

Q_{ji}	$y^*=0$	$y^*=1$	$y^*=2$	$y^*=3$
$\tilde{y}=0$	0.6	0	0	0.4
$\tilde{y}=1$	0.4	0.6	0	0
$\tilde{y}=2$	0	0.4	0.6	0
$\tilde{y}=3$	0	0	0.4	0.6

Q_{ji}	$y^*=0$	$y^*=1$	$y^*=2$	$y^*=3$
$\tilde{y}=0$	0.2	0	0.4	0.4
$\tilde{y}=1$	0.4	0.2	0	0.4
$\tilde{y}=2$	0.4	0.4	0.2	0
$\tilde{y}=3$	0	0.4	0.4	0.2

Q_{ji}	$y^*=0$	$y^*=1$	$y^*=2$	$y^*=3$
$\tilde{y}=0$	0.1	0.3	0.3	0.3
$\tilde{y}=1$	0.3	0.1	0.3	0.3
$\tilde{y}=2$	0.3	0.3	0.1	0.3
$\tilde{y}=3$	0.3	0.3	0.3	0.1

Figure 1: Example NTMs of different types of noisy labels with different noise rates ε.

Note that some works (Nguyen et al., 2020) assumed a sample is clean if the predicted label is the same as the annotated label $\widetilde{y} = \arg\max \hat{p}_\theta(\widetilde{y}|x)$. This is actually another form of describing small-loss assumption when selecting cross-entropy loss in classification.

3 Learning Behavior of DNNs Trained with Noisy Labels

Previous works empirically demonstrate that training DNNs with noisy labels contains two phases, including first *learning* patterns from the easy/clean samples and then *memorizing* the hard/noisy labels (Arpit et al., 2017). In this section, we explicitly formulate these two phases in Section 3.1. We characterize the learning behavior of DNNs in each phase in Section 3.2 and Section 3.3, respectively. The experimental evaluations of our theorems are shown in Section 3.4.

3.1 Formulate Training Process

Under the noise transition probabilities, the expectation $\mathbb{E}[\ell]$ of loss $\ell(\hat{p}_\theta(\widetilde{y}|x), \widetilde{y})$ equals:

$$\mathbb{E}[\ell] = \mathbb{E}[\ell(\hat{p}_\theta(\widetilde{y}|x), \widetilde{y})] = -\sum_i \sum_j q_{ji} \log \hat{p}_\theta(\widetilde{y} = j|y^* = i, x) \tag{2}$$

Since DNNs have capacity to memorize all training samples (Zhang et al., 2021), indicating $\mathbb{E}[\ell]$ will be degraded to 0 after sufficient training epochs, we model the optimization process as follows:

$$\mathbb{E}[\ell] = -\sum_i \sum_j q_{ji} \log \hat{p}_\theta(\widetilde{y} = j|y^* = i, x) \geq -\sum_i \sum_j q_{ji} \log q_{ji} \geq 0 \tag{3}$$

Based on Eqn.3, we divide the training process that optimizes $\mathbb{E}[\ell] \to 0$ into two phases, $\mathbb{E}[\ell^1]$ and $\mathbb{E}[\ell^2]$, where each phase corresponds to an inequality. Formally definitions are shown as follows.

Definition 6 (*The learning phase*): We define the training process that $\mathbb{E}[\ell^1] = \mathbb{E}[\ell] \to -\sum_i \sum_j q_{ji} \log q_{ji}$ as the learning phase. In this phase, DNNs are learning noise distribution.

Definition 7 (*The memorization phase*): We define the training process that $\mathbb{E}[\ell^2] = -\sum_i \sum_j q_{ji} \log q_{ji} \to 0$ as the memorization phase. In this phase, DNNs are overfitting to the noisy one-hot labels.

3.2 The Learning Phase: $\mathbb{E}[\ell^1] = \mathbb{E}[\ell] \to -\sum_i \sum_j q_{ji} \log q_{ji}$

We characterize the learning behavior of DNNs in the learning phase by proposing *Theorem 1*.

Theorem 1. *In the learning phase, minimizing $\mathbb{E}[\ell^1]$ will guide DNNs converging to the noise distributions until the outputs of the optimal model f_1^* equal to NTM when $\mathbb{E}[\ell^1] = -\sum_i \sum_j q_{ji} \log q_{ji}$.* (Detail proof is shown in Appendix A)

Remark 1. If we rewrite $\mathbb{E}[\ell^1]$ from the vector form as follows:

$$\mathbb{E}[\ell^1] = -\sum_i \sum_j q_{ji} \log \hat{p}_\theta(\widetilde{y}=j|y^*=i,x) = -\sum_i \sum_j q_{ji}(e^j)^T \log \hat{p}_\theta(\widetilde{y}|y^*=i,x) \tag{4}$$

Note that $\sum_j q_{ji}(e^j)^T = (Q_{*i})^T$, thus $\mathbb{E}[\ell^1]$ equals:

$$\mathbb{E}[\ell^1] = -\sum_i (Q_{*i})^T \log \hat{p}_\theta(\widetilde{y}|y^*=i,x) \tag{5}$$

This provides us a new insight into the learning phase as shown in Eqn.5, that is the i^{th} column of NTM Q_{*i} can be viewed as a *soft label* for the sample $x \in X_{y^*=i}$, indicating for both clean samples $x \in X_{\widetilde{y}=i,y^*=i}$ and noisy samples $x \in X_{\widetilde{y}=j,y^*=i}$, $j \neq i$, the outputs of DNNs $\hat{p}_\theta(\widetilde{y}|y^*=i,x)$ will consistently converge to the vector Q_{*i}. Experimental demonstration is shown in Section 3.4.

3.3 THE MEMORIZATION PHASE: $\mathbb{E}[\ell^2] = -\sum_i \sum_j q_{ji} \log q_{ji} \to 0$

Note that f_1^* in the learning phase $\mathbb{E}[\ell^1]$ is not the global optimization since the total loss $\mathbb{E}[\ell]$ will continually degrade to 0 when $\hat{p}_\theta(\widetilde{y}=j|y^*=i,x)=1, \forall i, \forall j$. We refer to this training process as the memorization phase and characterize the learning behavior of DNNs by proposing *Theorem 2*.

Theorem 2. *In the memorization phase, minimizing $\mathbb{E}[\ell^2]$ will force DNNs separately overfitting each subset $X_{\widetilde{y}=j,y^*=i} \in X_{y^*=i}$, $\forall i$, and the prediction of sample $x \in X_{\widetilde{y}=j,y^*=i}$ will converge to its one-hot noisy labels e^j until $\mathbb{E}[\ell^2]=0$.* (Detail proof is shown in Appendix A)

Theorem 2 is intuitively reasonable and is consistent with the memorization (overfitting) process that DNNs would over-confidently classify a sample to its noisy label by outputting a (near) one-hot prediction without considering the hidden true label.

3.4 EXPERIMENTS

Goal. We design experiments to evaluate *Theorem 1* and *Theorem 2*. Specifically, for the sample set $X_{y^*=i} = \sum_j X_{\widetilde{y}=j,y^*=i}$, $\forall i$, we demonstrate: (1) in the *learning phase*, DNNs converge to the i^{th} column of NTM Q_{*i} for both clean sample $x \in X_{\widetilde{y}=i,y^*=i}$ and noisy sample $x \in X_{\widetilde{y}=j,y^*=i}$; (2) in the *memorization phase*, DNNs separately overfit each subset $X_{\widetilde{y}=j,y^*=i}$ by guiding sample $x \in X_{\widetilde{y}=j,y^*=i}$ to converge to its one-hot noisy label e^j.

Datasets and models. We experiment on the CIFAR-10 dataset (Krizhevsky et al., 2009) and the MNIST datasets (LeCun et al., 1998). For CIFAR-10, we train a convolutional neural network RestNet-18 (He et al., 2016). For MNIST, we train a 3-layer MLP where each hidden layer contains 512 neurons. We train 200 epochs for both datasets. Detail training hyperparameters are shown in *Appendix B.1*.

Method. We directly track the outputs of DNNs from the start epoch to the end epoch. Specifically, for each sample set $X_{\widetilde{y}=j,y^*=i}$ in $X_{y^*=i}$, we average the softmax prediction vectors $\overline{P}_{ji} = \frac{\sum_{x \in X_{\widetilde{y}=j,y^*=i}} \hat{p}_\theta(\widetilde{y}|y^*=i,x)}{|X_{\widetilde{y}=j,y^*=i}|}$ and plot the trajectory images of $\overline{P}_{ji}$ during the entire training process.

To tackle the problem of plotting 10-dimensional outputs on CIFAR-10 and MNIST, we synthesized *2/3-Pariflip* noisy labels and simplify the problem into 2/3-D tracking by only tracing the non-zero dimensions in NTM. This is reasonable since the zero dimensions are irrelevant to the noisy labels. For example, for the *2-Pairflip* noisy labels, if the 1^{th} column of NTM is a 10-dimension vector $(0.6, 0.4, 0, \ldots)$, we only track the outputs in the first two dimensions. The training label $\widetilde{y}$ for the clean sample $x \in X_{\widetilde{y}=1,y^*=1}$ and the noisy sample $x \in X_{\widetilde{y}=2,y^*=1}$ can be simplified as $(1,0)$ and $(0,1)$, respectively.

Results. Figure 2 (left) shows the outputs trajectory of DNNs trained with *2-Pairflip* noisy labels on the CIFAR-10 dataset. We track the sample set $X_{y^*=1}$ and the noisy distribution of $X_{y^*=1}$ is $Q_{*1} = (0.4, 0.6)$, indicating $p(\widetilde{y}=1|y^*=1)=0.4$ and $p(\widetilde{y}=2|y^*=1)=0.6$. We center at Q_{*1}

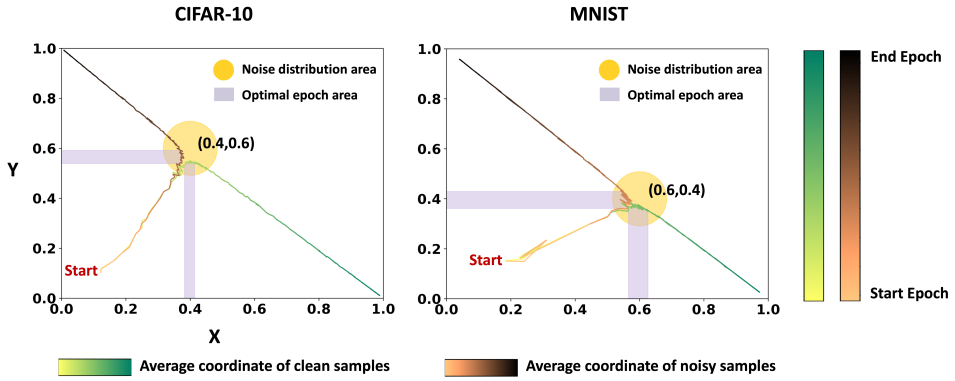


Figure 2: The trajectory of the softmax output values of DNNs trained with 2-*Pairflip* noisy labels. For the CIFAR-10 dataset, we track the sample set $X_{y^*=1}$ and the noise distribution is $Q_{*1} = (0.4, 0.6)$. For the MNIST dataset, we track the sample set $X_{y^*=2}$ and the noise distribution is $Q_{*2} = (0.6, 0.4)$. The training label $\widetilde{y}$ for the clean sample and the noisy sample is simplified as $(1, 0)$ and $(0, 1)$, respectively.

and draw a circle with a radius 0.1. We name this circle as the noise distribution area. Coordinate $(1, 0)$ and $(0, 1)$ is simplified as the training label for the clean sample $x \in X_{\widetilde{y}=1,y^*=1}$ and the noisy sample $x \in X_{\widetilde{y}=2,y^*=1}$, respectively. The gradual changing green line denotes the output trajectory $\overline{P}_{11}$ of the clean samples. The gradual changing red line denotes the output trajectory $\overline{P}_{21}$ of the noisy samples. We refer to the purple area as the optimal epoch area, which reflects the trajectory of ± 10 epochs of the approximated optimal model $\widetilde{f}_1^*$ in $\mathbb{E}[\ell^1]$.

As we can see, from the start epoch to the optimal epoch area, both clean samples $X_{\widetilde{y}=1,y^*=1}$ and noisy samples $X_{\widetilde{y}=2,y^*=1}$ have nearly the same predictions and together converge to the target noise distribution area. This is consistent with our defined learning phase as shown in *Theorem 1*. However, from the optimal epoch area to the end epoch, the outputs of $X_{\widetilde{y}=1,y^*=1}$ and $X_{\widetilde{y}=2,y^*=1}$ converge to their corresponding one-hot labels separately. This is consistent with our defined memorization phase as shown in *Theorem 2*.

Figure 2 (right) shows the consistent results on the MNIST dataset when we track the sample set $X_{y^*=2}$ in which the noise distribution is simplified as $Q_{*2} = (0.6, 0.4)$. More results of DNNs trained with *2-Pairflip* noisy labels on both datasets are shown in *Appendix B.2*.

We also experiment on *3-Pairflip* noisy labels. For CIFAR-10, the noise distribution of the sample set $X_{y^*=1}$ is simplified as $Q_{*1} = (0.5, 0.2, 0.3)$, indicating $p(\widetilde{y} = 1|y^* = 1) = 0.5$, $p(\widetilde{y} = 2|y^* = 1) = 0.2$ and $p(\widetilde{y} = 3|y^* = 1) = 0.3$. For MNIST, the noise distribution of the sample set $X_{y^*=2}$ is simplified as $Q_{*2} = (0.1, 0.6, 0.3)$, indicating $p(\widetilde{y} = 2|y^* = 2) = 0.1$, $p(\widetilde{y} = 3|y^* = 2) = 0.6$ and $p(\widetilde{y} = 4|y^* = 2) = 0.3$. The training label of the clean sample is simplified as $(1, 0, 0)$. The training label of the noisy sample is simplified as $(0, 1, 0)$ and $(0, 0, 1)$, respectively.

Output trajectories of DNNs trained with *3-Pairflip* noisy labels are shown in Figure 3 and we can see the consistent results. More results trained by *3-Pairflip* noisy labels are shown in *Appendix B.2*.

3.5 Summary

Unlike most of the previous works *qualitatively* describe that training DNNs with noisy labels contains a learning phase and a memorization phase, we *quantitatively* demonstrate by theorems and experiments that the outputs of DNNs first converge to the noise distribution in the learning phase and then overfit to the one-hot noisy labels in the memorization phase. Our comprehensive characterizing of the training process helps us provide new insights into two common challenges in learning with noisy labels: (1) NTM Estimation; (2) Robust training DNNs under noisy labels. We focus on these two challenges in Section 4 and Section 5, respectively.

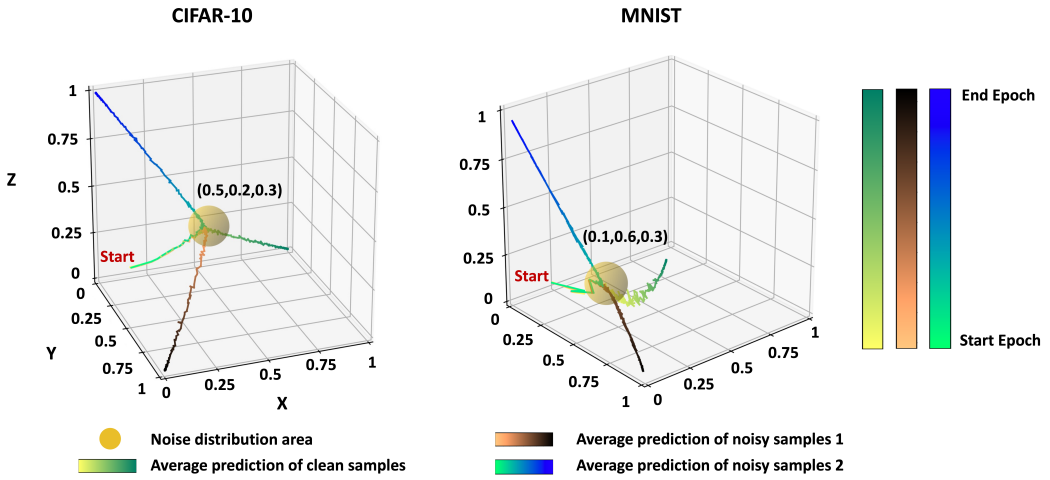


Figure 3: The trajectory of the softmax outputs of DNNs trained with 3-*Pairflip* noisy labels. For the CIFAR-10 dataset, we track the sample set $X_{y^*=1}$ and the noise distribution is $Q_{*1} = (0.5, 0.2, 0.3)$. For the MNIST dataset, we track the sample set $X_{y^*=2}$ and the noise distribution is $Q_{*2} = (0.1, 0.6, 0.3)$. The training label $\widetilde{y}$ for the clean sample is simplifies as $(1, 0, 0)$. The training label of noisy sample is simplified as $(0, 1, 0)$ and $(0, 0, 1)$, respectively.

4 Estimation of the Noise Transition Matrix

4.1 Two Simple Approaches to Estimate NTM

By applying a clean validation set, we propose two simple methods to estimate NTM directly from the outputs of DNNs. We name the first method as the Median Value Estimation (MV-E) which calculates the median value of the outputs based on the learning phase. We name the second method as the Statistics Confusion Matrix Estimation (SCM-E) which calculates the prediction confusion matrix based on the memorization phase. We first propose two theorems as follows.

Theorem 3. *In the learning phase, for the optimal model f_1^* when $\mathbb{E}[\ell^1] = -\sum_i \sum_j q_{ji} \log q_{ji}$, f_1^* can correctly predict the true label if sample is clean.* (Detail proof is shown in Appendix A)

Theorem 4. *In the memorization phase, for the perfect overfitting model f_2^* when $\mathbb{E}[\ell^2] = 0$, the prediction confusion matrix M of f_2^* equals NTM Q.* (Detail proof is shown in Appendix A)

Now we describe our estimation methods.

MV-E: Based on *Theorem 1* that the outputs of f_1^* in $\mathbb{E}[\ell^1]$ equal NTM, we can estimate NTM by finding a model $\widetilde{f}_1^*$ approximated to f_1^*. Based on *Theorem 3* that f_1^* can correctly predict the label if sample is clean, we use the optimal model $\widetilde{f}_1^*$ which has the best performance on validation set to approximate f_1^*. Formally, we calculate the median value of the softmax outputs $\hat{p}_\theta(\widetilde{y}|y^* = i, x)$ of $\widetilde{f}_1^*$ on validation sample set $X_{y^*=i}$ to estimate the i^{th} column of NTM.

SCM-E: Based on *Theorem 4*, we first train DNNs under sufficient epochs to ideally overfit the noisy training set by minimizing the loss (near) to 0. Then we directly use the prediction confusion matrix of DNNs on validation set to estimate NTM.

4.2 Experiments

We experiment on CIFAR-10 and MNIST datasets. We synthesized both K-*Pairflip* noisy labels and *Symmetry* noisy labels with different NTM on the training set. The test set is used for applying MV-E and SCM-E to estimate NTM. The models and the experiment configurations are the same as in Section 3.4. Example estimation results on MNIST are shown in Figure 4. As we can see, both

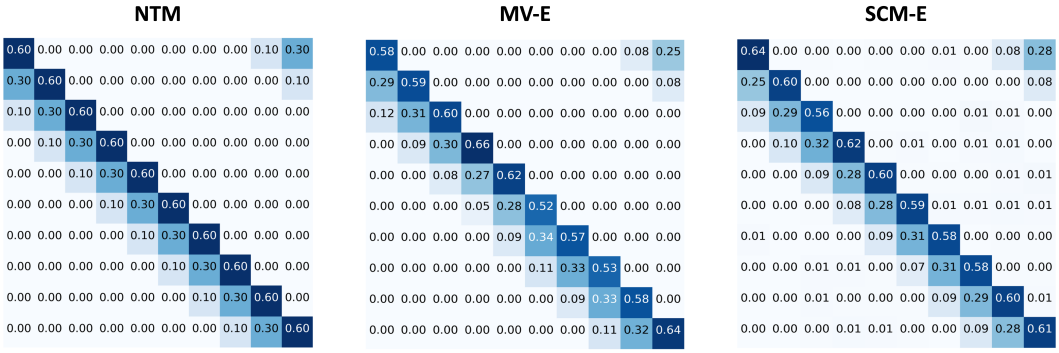


Figure 4: NTM estimation results of MV-E and SCM-E on the MNIST dataset.

MV-E and SCM-E can effectively estimate the target NTM. More estimation results are shown in *Appendix C.1*.

A major limitation of our estimation methods is that we need a clean validation set. Compared to SCM-E which needs a certain number of clean samples to calculate the confusion matrix, MV-E more relies on the outputs of DNNs. Based on this, we can build a tiny validation set where each class only consists of one clean sample, then we use MV-E to estimate NTM. This can sufficiently alleviate the requirement of clean samples and MV-E still can effectively estimate NTM. Detailed results are shown in Appendix C.2.

5 Robust Training DNNs under Noisy Labels

5.1 The Limitation of the Small-loss Assumption

How robust training DNNs under noisy labels is a challenging task. One of the most popular strategies is to isolate the gradient of noisy labels in back-propagation by filtering out noisy samples in each mini-batch data, typical methods include *Co-teaching* (Han et al., 2018b), *Co-teaching+* (Yu et al., 2019), *JoCoR* (Wei et al., 2020) and more (Jiang et al., 2018; Ren et al., 2018; Arazo et al., 2019; Nguyen et al., 2019). The basic idea is to assume the noisy samples have a larger loss than clean samples in the early training phase, deriving from the behavior that DNNs tend to learn simple patterns first (Arpit et al., 2017). We refer to this idea as the small-loss assumption and Algorithm 1 in *Appendix D.1* shows the general procedure of these methods.

However, we argue that the small-loss assumption has a critical limitation by proposing *Theorem 5*.

Theorem 5. *When using cross-entropy loss in image classification, the small-loss assumption holds if and only if when NTM Q is the column diagonally-maximum matrix.* (Detail proof is shown in Appendix A)

Theorem 5 provides a scenario that we cannot filter out the noisy samples by simply comparing the cross-entropy loss between samples, indicating existing methods will fail in image classification when NTM is not a column diagonally-maximum matrix e.g., *Co-teaching*. The experimental demonstrations are shown in Section 5.3.

5.2 Modify Models to Re-handle Noisy Labels

For the failure of the classification methods like *Co-teaching*, the limitation is only on the small-loss assumption rather than the whole training strategy. This provides us an opportunity to help these methods re-handle the noisy labels by only modifying the small-loss assumption.

Specifically, rather than assuming clean samples have smaller cross-entropy loss, we assume that a sample is clean when its predicted label $\hat{y}$ is the same as the annotated label $\widetilde{y}$, where the predicted label $\hat{y}$ is the maximum *cosine* similarity with the i^{th} column of NTM Q_{*i}:

Table 1: Comparison results of different models on the MNIST dataset

NTM	Noise rate	Co -teaching	Co-teaching*	Co -teaching+	Co-teaching+*	JoCoR	JoCoR*
Non -CDMM	Pair-60%	6.32	**95.59**	27.83	**82.53**	10.57	**94.61**
	Pair-80%	5.74	**95.43**	11.70	**90.72**	9.96	**96.20**
	Sym-99%	10.17	**73.66**	6.17	**69.14**	9.45	**87.06**
CDMM	Pair-40%	95.55	97.25	93.59	91.79	96.06	96.18
	Pair-45%	95.77	97.22	84.87	88.51	95.16	95.87
	Sym-40%	97.27	97.65	95.82	95.76	97.15	96.47
	Sym-80%	92.70	92.05	81.49	83.64	82.24	82.70

* CDMM: column diagonally-maximum matrix. Pair: *Pairflip* noisy labels, Sym: *Symmetry* noisy labels.

Table 2: Comparison results of different models on the CIFAR-10 dataset

NTM	Noise rate	Co -teaching	Co-teaching*	Co -teaching+	Co-teaching+*	JoCoR	JoCoR*
Non -CDMM	Pair-60%	3.41	**77.81**	24.18	**45.01**	6.62	**78.30**
	Pair-80%	5.99	**78.97**	17.46	**51.74**	5.64	**79.23**
	Sym-99%	9.99	**33.74**	9.78	**27.52**	7.43	**39.23**
CDMM	Pair-40%	74.86	77.12	48.12	46.97	71.45	75.18
	Pair-45%	72.07	77.59	42.35	42.83	65.01	73.96
	Sym-40%	79.10	77.36	55.66	56.07	79.64	77.08
	Sym-80%	31.18	33.69	34.59	36.84	22.51	30.12

* CDMM: column diagonally-maximum matrix. Pair: *Pairflip* noisy labels, Sym: *Symmetry* noisy labels.

$$\hat{y} = \arg\max_{i \in [c]} \frac{\hat{p}_\theta(\widetilde{y}|y^* = i, x) \cdot Q_{*i}}{||\hat{p}_\theta(\widetilde{y}|y^* = i, x)|| \times ||Q_{*i}||} \tag{6}$$

This idea is based on the *Remark 1* that the softmax outputs $\hat{p}_\theta(\widetilde{y}|y^* = i, x)$ will converge to the i^{th} column of NTM Q_{*i} in the learning phase, indicating $\hat{y} = i = \widetilde{y}$ for clean samples $x \in X_{\widetilde{y}=i,y^*=i}$ and $\hat{y} = i \neq \widetilde{y} = j$ for noisy samples $x \in X_{\widetilde{y}=j,y^*=i}$. Note that we also use Eqn.6 to predict labels during the testing process. One limitation of this method is it requires knowing Q. If Q is unknowing, we can use the method MV-E or SCM-E proposed in Section 4 to pre-estimate.

Overall, by substituting the small-loss assumption with Eqn.6, methods like *Co-teaching* can re-handle the noisy labels when NTM is not a column diagonally-maximum matrix. The overall training strategy is shown in Algorithm 2 in *Appendix D.1*. We also give concrete pseudo-codes of modifying strategies on *Co-teaching* and *JoCoR* in *Appendix D.1*.

5.3 Experiments

Goal. We demonstrate when using cross-entropy loss in image classification, the methods designed based on the small-loss assumption will fail when NTM is *not* a column diagonally-maximum matrix. However, by substituting our proposed method, these methods can tackle this problem.

Dataset. We experiment on the MNIST dataset and CIFAR-10 dataset. We split 20% training data as the validation set for pre-estimating NTM. Like previous works, we synthesize *2-Pairflip* and *Symmetry* noisy labels with different NTM on the remaining 80% training data.

Models. We select three typical DNNs including *Co-teaching*, *Co-teaching+* and *JoCoR*. We rebuild these DNNs from their public official codes. All models are based on the small-loss assumption and filtering out large cross-entropy loss samples in the training process. The differences between three models are summarized in *Appendix D.2*.

Results. We use superscript "*" on each model to denote the modified version based on our proposed method. The testing accuracy on two datasets is shown in Table 1 and Table 2, respectively.

As we can see, when NTM is *not* a column diagonally-maximum matrix, all three methods fail to classify images since most of the accuracy is nearly 10%, which equals the probability of random inference under a 10-class classification. However, with the same NTM, our proposed method help all models significantly improve their classification accuracy.

When NTM is a column diagonally-maximum matrix, the results of our proposed method are similar to the models that are using small-loss assumption. This is reasonable since the predicted clean samples from Eqn.6 also have a smaller loss compared to the noisy samples.

In summary, we demonstrate the small-loss assumption has a critical limitation. The DNNs designed based on this assumption fail to classify images when NTM is *not* a column diagonally-maximum matrix. However, our proposed method tackles this problem and helps these models recover the classification ability under label noises.

6 Related work

Learning behavior of DNNs trained with noisy labels. Zhang et al. (2021) demonstrated that DNNs have the ability to memorize the whole training dataset under randomization labels. Arpit et al. (2017) further demonstrated that DNNs prioritize learning patterns before memorizing noisy labels. This phenomenon inspires substantial research to characterize the learning behavior of DNNs trained with noisy labels. For example, Xu & Raginsky (2017) proposed upper bounds on generalization errors in terms of mutual information between input and output. Chen et al. (2019) shown empirically that generalization accuracy can be quantitatively characterized in terms of noise ratio in datasets. Harutyunyan et al. (2020) quantified the memorization information in DNNs by using Shannon mutual information between weights and labels. Luo et al. (2022) demonstrate that DNNs learn spatial structure information in semantic segmentation under pixel-level label noises. Compared to these studies, we characterize the learning behavior of DNNs by explicit formulating the training process in image classification.

Estimation of NTM. A number of methods are proposed to estimate NTM, including modeling extra layers(Sukhbaatar et al., 2014; Jindal et al., 2016), building separate netwroks(Xiao et al., 2015; Han et al., 2018a; Yao et al., 2018), explicit calculating (Bekker & Goldberger, 2016; Goldberger & Ben-Reuven, 2017; Patrini et al., 2017), using convergence criterion (Scott, 2015), and predicted probability ratios (Northcutt et al., 2017). Unlike previous works, we provide two estimation methods which are based on the outputs of DNNs.

Robust training DNNs with noisy labels. Based on the small-loss assumption, Han et al. (2018b) proposed *Co-teahcing* by training two DNNs simultaneously and letting them teach each other by filtering out noisy labels. Yu et al. (2019) further developed *Co-teaching* into *Co-teaching+* by adding an updating rule "Update by disagreement" (Malach & Shalev-Shwartz, 2017). Wei et al. (2020) also proposed two DNNs to teach each other by reducing the prediction diversity of two networks. Other similar methods including Vahdat (2017); Yi & Wu (2019); Li et al. (2020); Nguyen et al. (2020). However, in this paper, we expose the limitation of the small-loss assumption and demonstrate that these models fail when NTM is not a column diagonally-maximum matrix.

7 Conclusion

In this work, we explicitly model the training process of DNNs under noisy labels into two phases, the learning phase in which the outputs of DNNs converge to the noise distribution, and the memorization phase in which the outputs of DNNs converge to the noisy one-hot labels. The characteristic of these two phases inspires us to propose two simple methods to estimate the NTM directly from the outputs of DNNs. It also helps us resolve a pitfall of existing methods for robust training under noisy labels based on the small-loss assumption. The limitation of our proposed robust training strategy is that we need pre-estimate NTM, and NTM estimation requires a clean validation set. Despite these limitations, the learning behavior of DNNs characterized in this study provides new insight into how DNNs learn from noisy labels in image classification.

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A THEOREMS AND PROOFS

In this section, we restate the theorems proposed in our paper and provide the corresponding proofs.

Theorem 1. *In the learning phase, minimizing $\mathbb{E}[\ell^1]$ will guide DNNs converging to the noise distributions until the outputs of the optimal model f_1^* equal to NTM when $\mathbb{E}[\ell^1] = -\sum_i \sum_j q_{ji} \log q_{ji}$.*

Proof. Since $\mathbb{E}[\ell^1]$ is monotonic decreasing and with equality only when $\hat{p}_\theta(\widetilde{y} = j|y^* = i, x) = q_{ji}$. Thus, minimizing $\mathbb{E}[\ell^1]$ will force model converging to the noisy distributions and the outputs of f_1^* equal to NTM. □

Theorem 2. *In the memorization phase, minimizing $\mathbb{E}[\ell^2]$ will force DNNs separately overfitting each subset $X_{\widetilde{y}=j,y^*=i} \in X_{y^*=i}$, $\forall i$, and the prediction of sample $x \in X_{\widetilde{y}=j,y^*=i}$ will converge to its one-hot noisy labels e^j until $\mathbb{E}[\ell^2] = 0$.*

Proof. Note that $-\sum_i \sum_j q_{ji} \log 1 = 0$, the memorization phase can be modified as:

$$\mathbb{E}[\ell^2] = -\sum_i \sum_j q_{ji} \log \hat{p}_\theta(\widetilde{y} = j|y^* = i, x) \geq -\sum_i \sum_j q_{ji} \log 1 \tag{7}$$

where $\hat{p}_\theta(\widetilde{y} = j|y^* = i, x) = q_{ji}$. The $\mathbb{E}[\ell^2]$ is monotonic decreasing and the prediction of the overfitting model f_2^* satisfy $\hat{p}_\theta(\widetilde{y} = j|y^* = i, x) = 1$. Note that if f_2^* outputs the same softmax predictions $\hat{p}_\theta(\widetilde{y}|y^* = i, x)$ for all samples $x \in X_{y^*=i}$, it will be paradoxical to the definition of the probability distribution because $\sum_j \hat{p}_\theta(\widetilde{y} = j|y^* = i, x) > 1$. However, if we rewrite the memorization phase after divide $X_{y^*=i}$ into subset $X_{\widetilde{y}=j,y^*=i} \in X_{y^*=i}$, $j \in [c]$:

$$\mathbb{E}[\ell^2] = -\sum_i \sum_j q_{ji} \cdot [(e^j)^T \log \hat{p}_\theta(\widetilde{y}|y^* = i, x \in X_{\widetilde{y}=j,y^*=i})] \geq -\sum_i \sum_j q_{ji} \cdot [(e^j)^T \log e^j] \tag{8}$$

Eqn.8 guarantees the softmax output of f_2^* is a probability distribution and the equality holds only when $\hat{p}_\theta(\widetilde{y}|y^* = i, x \in X_{\widetilde{y}=j,y^*=i}) = e^j$, indicating f_2^* separately fitting each subset $X_{\widetilde{y}=j,y^*=i}$ and the prediction of sample $x \in X_{\widetilde{y}=j,y^*=i}$ converges to the one-hot noisy labels e^j. □

Theorem 3. *In the learning phase, for the optimal model f_1^* when $\mathbb{E}[\ell^1] = -\sum_i \sum_j q_{ji} \log q_{ji}$, f_1^* can correctly predict the true label if sample is clean.*

Proof. Based on *Theorem 1* that the output of f_1^* equals NTM $\hat{p}_\theta(\widetilde{y} = j|y^* = i) = q_{ji}$, we modify equality as $\hat{p}_\theta(\widetilde{y} = j|y^* = i) = q_{ji} \cdot \hat{p}_\theta(y^* = i|x) = q_{ji}$, which means $\hat{p}_\theta(y^* = i|x) = 1$. Thus, f_1^* can correctly predict the true label if sample is clean. □

Theorem 4. *In the memorization phase, for the perfect overfitting model f_2^* when $\mathbb{E}[\ell^2] = 0$, the prediction confusion matrix M of f_2^* equals NTM Q.*

Proof. Based on *Theorem 2*, the outputs of f_2^* satisfy $\hat{p}_\theta(\widetilde{y} = j|y^* = i, x) = 1$ only when sample $x \in X_{\widetilde{y}=j,y^*=i}$, . Thus, the m_{ji} in M is equals the q_{ji} in Q:

$$m_{ji} = \frac{\sum_{x \in X_{y^*=i}} \hat{p}_\theta(\widetilde{y} = j | y^* = i, x)}{|X_{y^*=i}|} = \frac{|X_{\widetilde{y}=j, y^*=i}|}{|X_{y^*=i}|} = q_{ji} \tag{9}$$

Thus $M = Q$. □

Theorem 5. *When using cross-entropy loss in image classification, the small-loss assumption holds if and only if when NTM Q is the column diagonally-maximum matrix.*

Proof. If Q is *not* a column diagonal-maximum matrix, there exists $q_{ii} \leq q_{ji}$. Based on the *Theorem 1* that DNNs will converge to NTM in the learning process, the predictions of DNNs satisfy $\hat{p}_\theta(\widetilde{y} = i | y^* = i, x) \leq \hat{p}_\theta(\widetilde{y} = j | y^* = i, x)$ and the cross-entropy loss $-\log \hat{p}_\theta(\widetilde{y} = i | y^* = i, x) \geq -\log \hat{p}_\theta(\widetilde{y} = j | y^* = i, x)$, indicating the clean sample has a larger loss than the noisy sample and this is contradictory to the small-loss assumption. □

B Extra prediction trajectory of DNNs

B.1 Training hyperparameters

For training ResNet-18 on CIFAR-10, we set batchsize as 2048. The learning rate is 0.1 and the weight decay is 0.0001. For training 3-layer MLP on MNIST, we set batchsize as 8192. The learning rate is 0.3 and the weight decay is 0.00001. The momentum is set as 0.9 for both datasets.

B.2 2-D and 3-D tracing results

Figure 5 show the prediction trajectory of DNNs trained with three different 2-*Pairflip* noisy labels on the MNIST dataset. The noise distributions Q_{*2} equal $(0.8, 0.2)$, $(0.4, 0.6)$ and $(0.2, 0.8)$, respectively.

Figure 6 show the prediction trajectory of DNNs trained with three different 2-*Pairflip* noisy labels on the CIRAR-10 dataset. The noise distributions Q_{*1} equal $(0.8, 0.2)$, $(0.6, 0.4)$ and $(0.2, 0.8)$, respectively.

Figure 7 show the prediction trajectory of DNNs trained with two different 3-*Pairflip* noisy labels on the MNIST dataset. The noise distributions Q_{*2} equal $(0.6, 0.3, 0.1)$ and $(0.1, 0.6, 0.3)$, respectively.

Figure 8 show the prediction trajectory of DNNs trained with two different 3-*Pairflip* noisy labels on the CIFAR-10 dataset. The noise distributions Q_{*1} equal $(0.6, 0.3, 0.1)$ and $(0.1, 0.6, 0.3)$, respectively.

As we can see, all experimental results show the consistent phenomenon that the outputs of DNNs first converge to the noise distributions together and then overfit to the noisy one-hot labels separately.

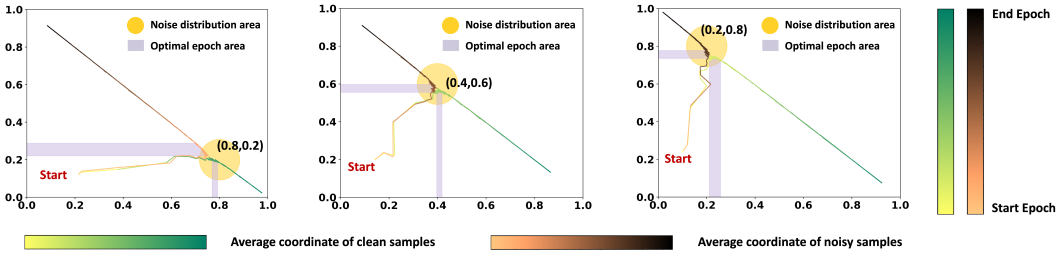

Figure 5: The trajectory of the softmax predictions of DNNs trained with 2-*Pairflip* noisy labels on the MNIST dataset.

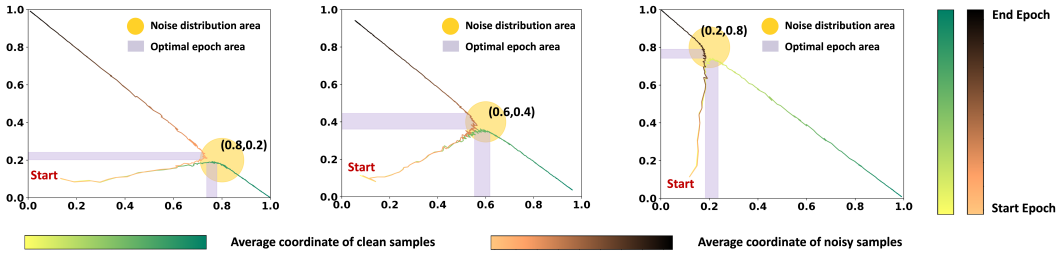

Figure 6: The trajectory of the softmax predictions of DNNs trained with 2-*Pairflip* noisy labels on the CIFAR-10 dataset.

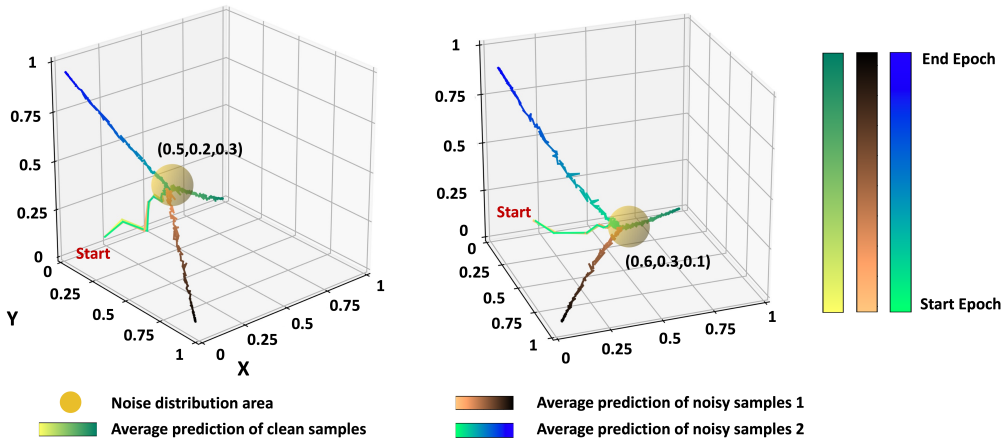

Figure 7: The trajectory of the softmax predictions of DNNs trained with 3-*Pairflip* noisy labels on the MNIST dataset.

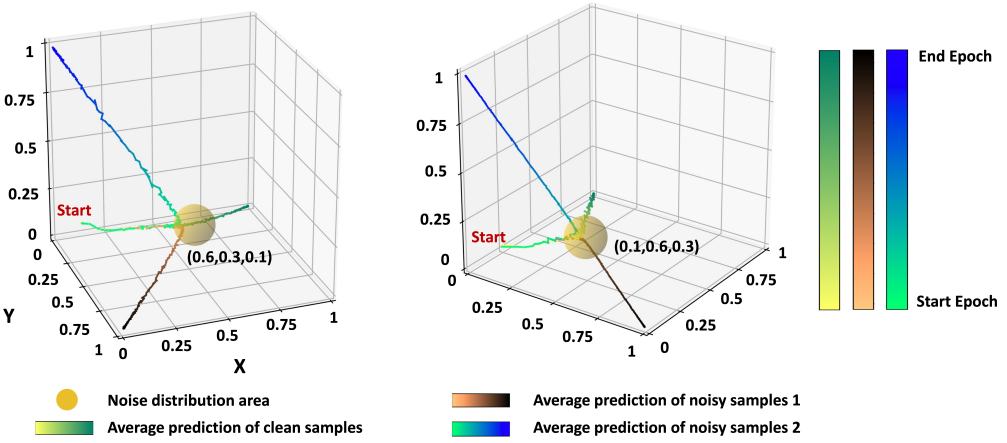

Figure 8: The trajectory of the softmax predictions of DNNs trained with 3-*Pairflip* noisy labels on the CIFAR-10 dataset.

C RESULTS OF NTM ESTIMATION

C.1 EVALUATE ON FULL VALIDATION SET

Figure 9 and Figure 10 shows more estimation results of NTM on the MNIST and CIFAR-10 datasets, respectively. Both datasets include *PairFlip* noisy labels and *Symmetry* noisy labels. As we can see, our methods MV-E and SCM-E can effective estimate the hidden noise distribution NTM.

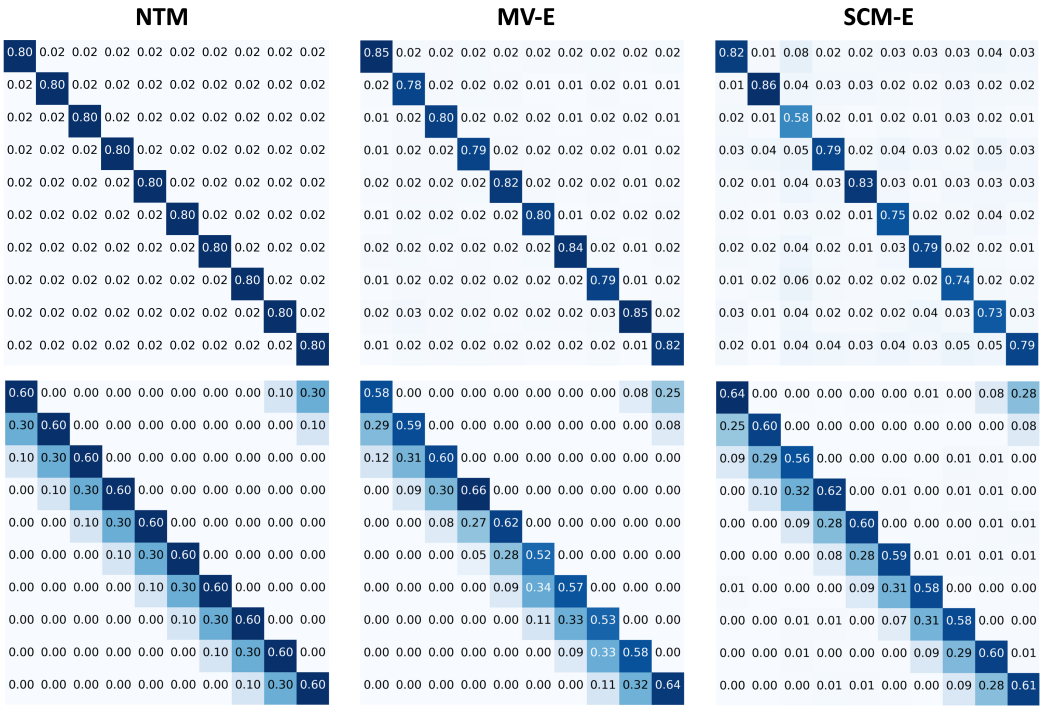

Figure 9: NTM estimation results of MV-E and SCM-E on the MNIST dataset.

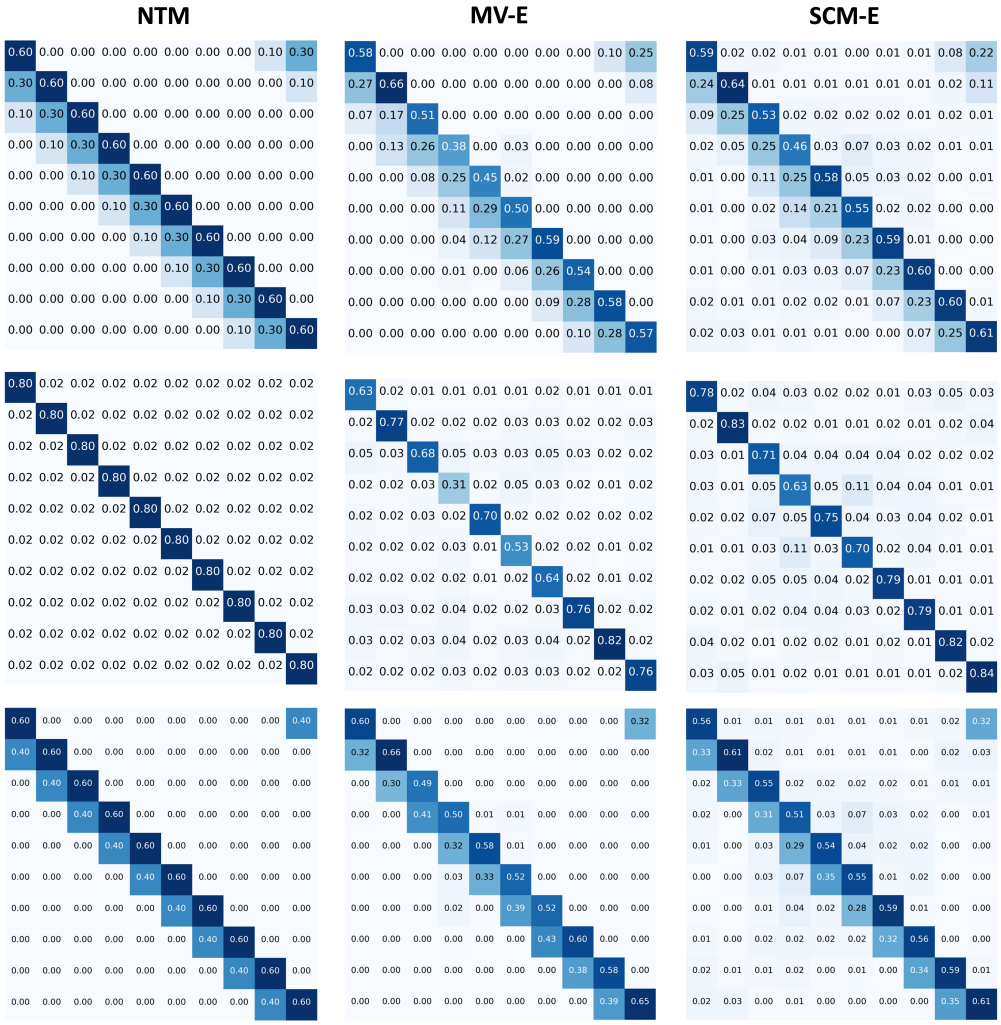

Figure 10: NTM estimation results of MV-E and SCM-E on the CIFAR-10 dataset.

C.2 EVALUATE ON TINY VALIDATION SET

Figure 11 shows the estimation results of NTM by using MV-E on tiny validation set, where each class only have one clean sample.

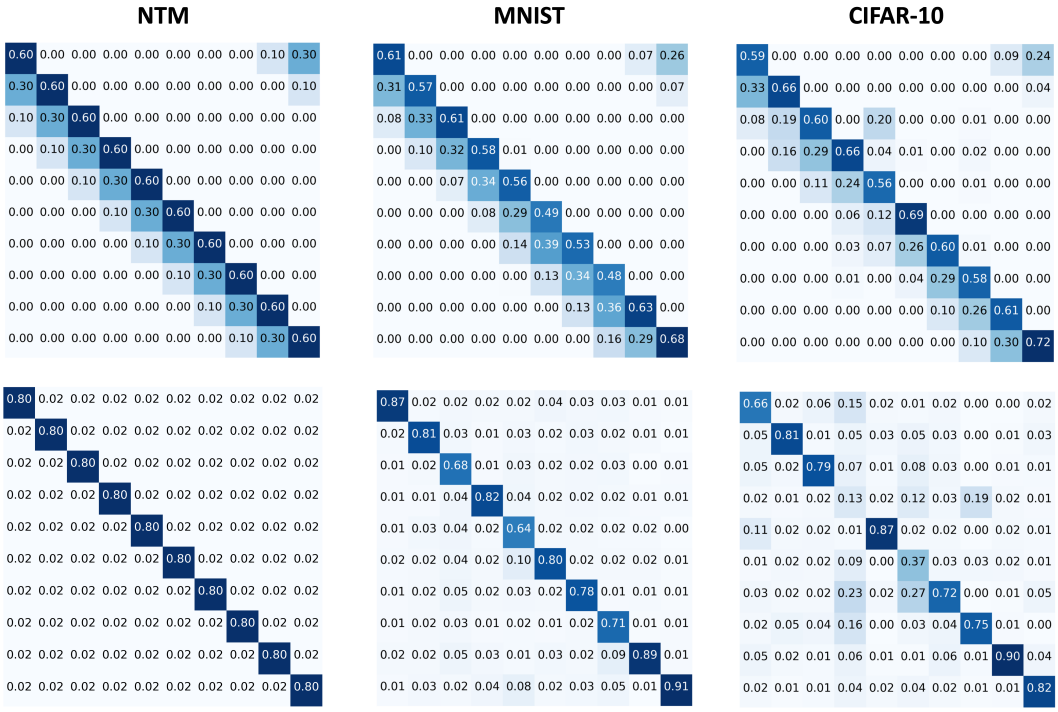

Figure 11: NTM estimation results of MV-E on the MNIST and CIFAR-10 dataset.

D EVALUATION OF THE SMALL-LOSS ASSUMPTION

D.1 PSEUDO CODE OF DNNS TRAINED WITH NOISY LABELS

Algorithm 1 provides the general procedure of DNNs trained with noisy labels. The critical step is to filter out the noisy samples by assuming that clean samples have a smaller loss, which we name as the small-loss assumption.

Algorithm 2 provides our improved version based on Algorithm 1. We substitute the small-loss assumption by viewing the sample as clean if $y = \arg\max_i CS(f(x)||Q_{*i})$.

Algorithm 3 and *Algorithm 4* give concrete examples to modified the *Co-teaching* and the *JoCoR*, respectively.

Algorithm 1 General procedure of robust training with noisy labels based on the small-loss assumption

Inputs: Network f, noisy label y, loss function ℓ, epoch N, threshold t;

1: **for** $i = 1, \cdots, N$ **do**
2: draw a mini-batch $\overline{D}$;
3: the sample $x \in \overline{D}$ is clean ***if***: $\ell(f(x), y) < t$ ***or*** $y = \arg\max f(x)$;
4: update network using clean samples;
5: **end for**

Algorithm 2 Improved general procedure of robust training with noisy labels

Inputs: Network f, noisy label y, NTM Q, cosine similarity CS, loss function ℓ, epoch n;

if Q is unknown **then**
  predict Q with MV-E or SCM-E;
else
  for $i = 1, \cdots, N$ **do**
    draw a mini-batch $\overline{D}$;
    the sample $x \in \overline{D}$ is clean ***if***: $y = \arg\max_i CS(f(x)||Q_{*i})$;
    update network using clean samples;
  end for
end if

Algorithm 3 Improved *Co-teaching* Algorithm

Inputs: w_f and w_g, learning rate η, NTM Q, cosine similarity CS, fixed τ, epoch T_k and T_{max}, iteration N_{max};

if Q is unknown **then**
  predict Q with MV-E or SCM-E;
else
  for $T = 1, \cdots, T_{max}$ **do**
    Shuffle training set D;
    for $N = 1, \cdots, N_{max}$ **do**
      Fetch mini-batch $\overline{D}$ from D;
      Obtain $\overline{D}_f = \arg\max_{D':|D'|\geq R(T)|\overline{D}|} CS_{x\in\overline{D}}(f(x)||Q_{*,i})$; //Sample $R(T)\%$ clean instances
      Obtain $\overline{D}_g = \arg\max_{D':|D'|\geq R(T)|\overline{D}|} CS_{x\in\overline{D}}(g(x)||Q_{*,i})$; //Sample $R(T)\%$ clean instances
      Update $w_f = w_f - \eta\nabla\ell(f, \overline{D}_f)$;
      Update $w_g = w_g - \eta\nabla\ell(g, \overline{D}_g)$;
    end for
    Update $R(T) = 1 - \min\{\frac{T}{T_k}\tau, \tau\}$;
  end for
  Output: w_f and w_g.
end if

Algorithm 4 Improved *JoCoR* Algorithm

Inputs: Network f with $\Theta = \{\Theta_1, \Theta_2\}$, learning rate η, NTM Q, cosine similarity CS, fixed τ, epoch T_k and T_{max}, iteration I_{max};

if Q is unknown **then**
  predict Q with MV-E or SCM-E;
else
  for $t = 1, \cdots, T_{max}$ **do**
    Shuffle training set D;
    for $n = 1, \cdots, I_{max}$ **do**
      Fetch mini-batch D_n from D;
      $p_1 = f(x, \Theta_1), \forall x \in D_n$;
      $p_1 = f(x, \Theta_1), \forall x \in D_n$;
      Obtain $\widetilde{D}_n = \arg\max_{D':|D'|\geq R(T)|D_n|} CS_{x\in D_n}(f(x)||Q_{*,i})$; //Sample $R(T)\%$ clean instances
      Calculate the joint loss $L = (1-\lambda) \times \ell(f, \widetilde{D}_n) + \lambda \times (KL(p_1||p_2) + KL(p_2||p_1))$;
      Update $\Theta = \Theta - \eta \nabla L$;
    end for
    Update $R(T) = 1 - \min\{\frac{T}{T_k}\tau, \tau\}$;
  end for
  Output: Θ_1 and Θ_2.
end if

D.2 Comparison of the state-of-the-art models

We compare the state-of-the-art models from the following techniques:

- *Small loss:* regarding small-loss samples as clean samples, which is based on the *small-loss* assumption;
- *Double classifiers:* training two classifiers simultaneously;
- *Cross update:* updating parameters in a cross manner instead of a parallel manner;
- *Joint training:* training two classifiers with a joint loss;
- *Disagreement:* updating two classifiers on disagreed examples during the entire training epochs;
- *Agreement:* maximizing the agreement of two classifiers by regularization during the whole training epochs.

Table 3: Difference between the state-of-the-art models

	Co-teaching	Co-teaching+	JoCoR
Small loss	✓	✓	✓
Double classifiers	✓	✓	✓
Cross update	✓	✓	
Joint training			✓
Disagreement		✓	
Agreement			✓